

Tutorial Sheet – 2

Course: B. Tech (CSE)

Year/Semester: 3rd, VI Sem

Session: 2023-24

Subject Name & Code: Theory of Automata & Formal Languages (BCSC0011)

Max. Marks:

Time allowed: 50 Min.

Note/Instruction (If any)

- Q1. Which of the following sub-strings will the following notation result? Initial state is Q0 and final state is Q4.

State	0	1
Q0-	Q1	Q4
Q1	Q4	Q2
Q2	Q3	Q3
Q3	Q2	Q2
Q4+	Q4	Q4

a) 0101011 b) 0101010 c) 010100 d) 100001

- Q2. Construct a DFA equivalent to an NFA whose transition table is defined:

state	a	b
q0-	q1, q3	q2, q3
q1	q1	q3
q2	q3	q2
q3+	-	-

Q3. From the given table, $\delta^*(q_0, 011) = ?$

State	0	1
-q0	{q0}	{q0, q1}
q1	{q2}	{q2}
q2	{q3}	{q3}
q4+	--	--

Q4. Construct a DFA equivalent to an NFA whose transition table is defined:

state	a	b
q0-	q0, q3	q0, q1
q1	-	q2
q2	q2	q2
q3	q4	-
q4+	q4	q4

Q5. Suppose Language L is given as:

$L = \{x \in \{a, b\}^* \mid x \text{ contains aba as its substring}\}$

Find the difference of transitions made in constructing a DFA and an equivalent NFA?